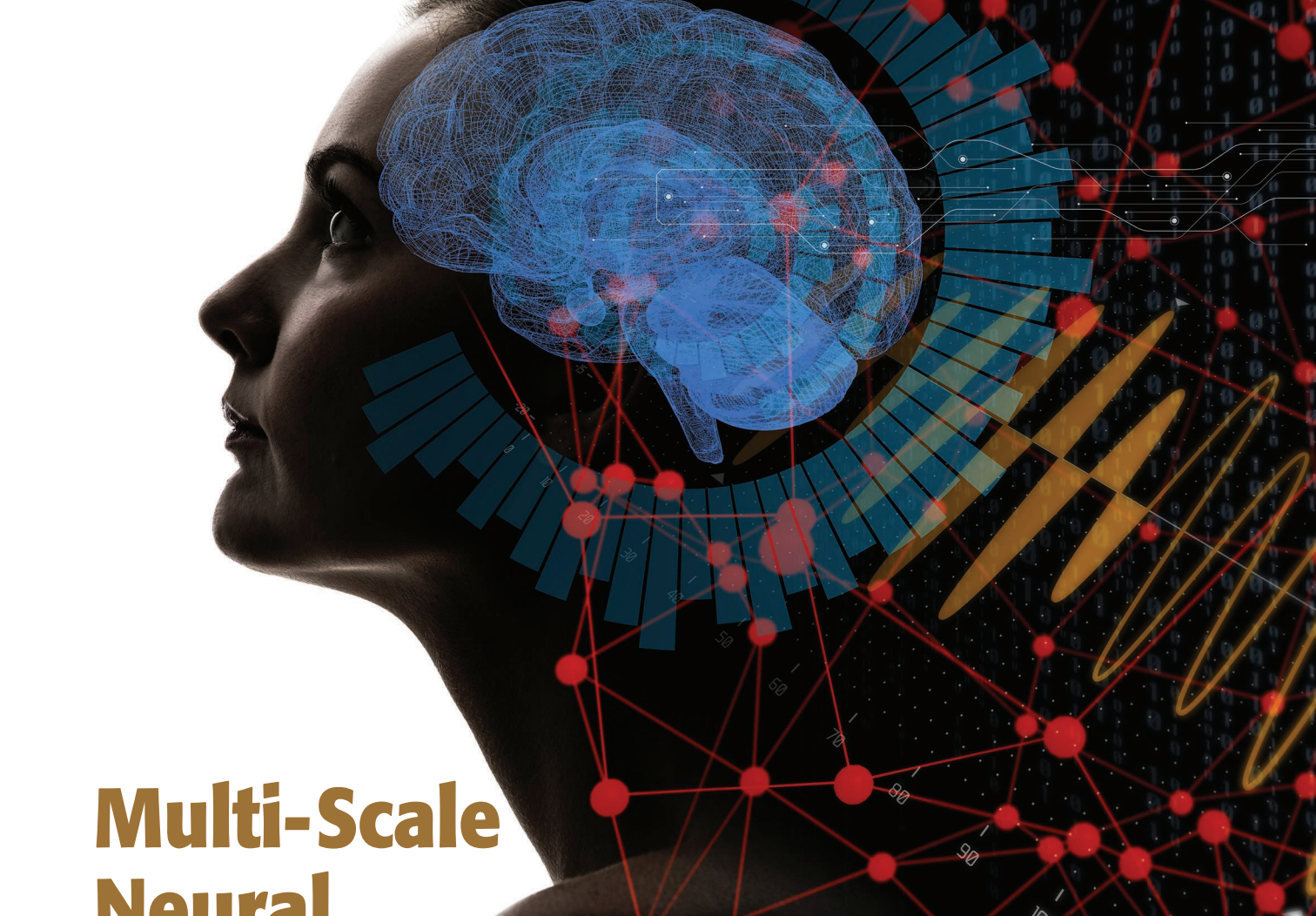




Multi-Scale Neural Network for EEG Representation Learning in BCI

©SHUTTERSTOCK.COM/METAMORWORKS

Wonjun Ko, Eunjin Jeon,
Seungwoo Jeong, and
Heung-Il Suk
Korea University, KOREA

Abstract—Recent advances in deep learning have had a methodological and practical impact on brain–computer interface (BCI) research. Among the various deep network architectures, convolutional neural networks (CNNs) have been well suited for *spatio-spectral-temporal* electroencephalogram (EEG) signal representation learning. Most of the existing CNN-based methods described in the literature extract features at a sequential level of abstraction with repetitive nonlinear operations and involve densely connected layers for classification. However, studies in neurophysiology have revealed that EEG signals

carry information in different ranges of frequency components. To better reflect these *multi-frequency* properties in EEGs, we propose a novel deep *multi-scale neural network* that discovers feature representations in multiple frequency/time ranges and extracts relationships among electrodes, *i.e.*, spatial representations, for subject intention/condition identification. Furthermore, by completely representing EEG signals with spatio-spectral-temporal information, the proposed method can be utilized for diverse paradigms in both active and passive BCIs, contrary to existing methods that are primarily focused on single-paradigm BCIs. To demonstrate the validity of our

proposed method, we conducted experiments on various paradigms of active/passive BCI datasets. Our experimental results demonstrated that the proposed method achieved performance improvements when judged against comparable state-of-the-art methods. Additionally, we analyzed the proposed method using different techniques, such as PSD curves and relevance score inspection to validate the multi-scale EEG signal information capturing ability, activation pattern maps for investigating the learned spatial filters, and t-SNE plotting for visualizing represented features. Finally, we also demonstrated our method's application to real-world problems. Based on our experimental results and analyses, we believe that the proposed multi-scale neural network can be useful for various BCI paradigms, as a starting model or as a backbone network in any new BCI experiments.

I. Introduction

Brain-computer interface (BCI) [1], [2] is an emerging technology that enables a communication pathway between a user and an external device (e.g., a computer) through the acquisition and analysis of brain signals. These signals are then translated into commands that are understood by a device, such as a computer. Owing to its practicality, electroencephalogram (EEG)-based non-invasive BCIs are widely used [1]–[5]. Earlier, Aricò *et al.* [6] categorized user-centered BCIs into two types, active/reactive and passive BCIs. In this paper, our focus is not only on active BCIs but also on passive BCIs. Generally, two types of brain signals such as *evoked* and *spontaneous* EEG are primarily considered for active/reactive BCIs [7]. Evoked BCIs exploit unintentional electrical potentials reacting to external or internal stimuli. Examples of evoked BCIs include steady-state visually evoked potentials (SSVEP) [8], [9] and event-related potentials such as P300 [8], [10]. Additionally, spontaneous BCIs use an internal cognitive process such as event related desynchronization and event related synchronization (ERD/ERS) in sensorimotor rhythms, e.g., motor imagery (MI) [4], [11] induced by imagining movements in addition to physical movement. Well-known examples of passive BCIs include the use of sleep/drowsy EEG signals for

sleep stage classification or identifying mental fatigue to alert a driver of a dangerous situation and seizure EEG patterns for onset detection to provide the patient with a warning of a potential seizure.

Generally, machine learning-based BCIs consist of five main processing stages [4]: (i) an EEG signal acquisition phase based on each paradigm, (ii) signal preprocessing (e.g., channel selection and band-pass filtering), (iii) feature representation learning, (iv) classifier learning, and finally (v) a feedback stage. Basically, most machine learning-based BCI methods follow these processes, however, these methods need specific modification to classify a user's intention/condition for each different paradigm [4]. In other words, machine learning-based methods need to have *prior* knowledge of different EEG paradigms [1], [4], [8], [9], [12]. Therefore, conventional machine learning-based BCIs have discovered EEG representations through extremely specialized approaches, e.g., a common spatial pattern (CSP) [1] or its variants [11], [13] for MI signals and a canonical correlation analysis (CCA) [9] for SSVEP signals decoding.

While hand-crafted feature representation learning has a pivotal role in a conventional machine learning framework [1], [9], [14]–[16], deep learning-based representation has had remarkable results in the BCI community [3], [4], [7], [17]. These deep learning-based methods have integrated a feature extraction step with a classifier learning step such that those steps are jointly optimized, thereby improving performance. Among various deep learning methods, convolutional neural networks (CNNs) have the advantage [4], [18], [19] of maintaining the structural and configurational information in the original data. In this respect, developing a novel CNN architecture for EEG signal representation has taken a center stage in the BCI studies [3], [18], [20]–[27].

However, some challenges still remain. First, existing CNN-based methods [3], [18], [20], [21], [25], [26], [28] are mostly comprised of stacked convolutional layers. In other words, those existing methods extract features sequentially. But, ignoring multiple ranges of spectral-temporal features can cause a critical problem because EEG signal features for different subjects [29], paradigms [4], and types [6] are found in diverse ranges. For example, Fig. 1 depicts two different subjects' MI EEG power spectral density (PSD) curves. Clearly, these two plots have different distributions from each other even though these PSDs are estimated by the same task. Therefore, it is important to capture multi-scale spectral information in EEGs for *general use* in BCI, i.e., a generic method applicable to various types of BCIs.

In addition, those stacked CNN-based methods [3], [18], [25], [26] have numerous trainable parameters, thus requiring large amounts of training samples, whereas BCIs generally acquire a limited number of EEG trials [29]. Therefore, generalizing conventional stacked CNN-based methods in BCI is quite difficult because deep learning is a *data-hungry* problem, i.e., rarely generalized with a lack of data.

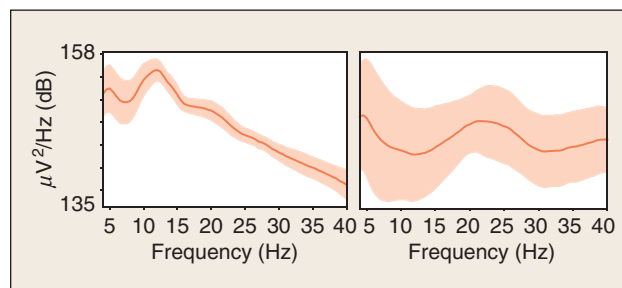


FIGURE 1 Power spectral density (PSD) curves of two different subjects' MI EEG samples. The solid red line denotes the mean PSD and the shaded region exhibits the standard deviation for all trials. Clearly, these two different subjects show quite different PSD patterns for the same paradigm. Note that two PSD plots share the y-axis label.

Finally, interpreting a learned stacked CNN from a neuro-physiologically appropriate standpoint [30] is quite complicated because the CNN identifies complex patterns of data in latent space, making a direct explanation difficult.

In this study, we propose a novel deep learning-based BCI method to mitigate the previously discussed difficulties. We conduct exhaustive experiments on various EEG paradigms and also perform diverse analyses. According to our observations, the proposed deep neural network achieved comparable performances in various paradigms with reasonable neurophysiological interpretations. In this regard, we believe that the proposed network can be used as a starting model and/or a backbone network for any new BCI protocols, experiments, and paradigms. The main contributions of our study are as follows:

- ❑ First, we propose a novel CNN architecture that is applicable independently from the input paradigm or types of EEG, and can represent multi-scale spatio-spectral-temporal features.
- ❑ Second, the proposed method achieved reasonable performance on five different datasets for four different paradigms (two for active BCIs and two for passive BCIs). The proposed method outperformed or was similar to state-of-the-art linear and deep learning methods, which were individually designed for each specific paradigm.
- ❑ Third, the proposed method showed robust performance on three different paradigms (MI, drowsy, and seizure) without specific tuning of network architecture, *e.g.*, kernel sizes.
- ❑ Lastly, we analyze the proposed network using a variety of techniques.

The rest of this paper is organized as follows: Section II reviews previous research on various EEG representation learning via linear model-based or deep learning-based methods. In Section III, we propose a novel and compact deep CNN that classifies multi-paradigm EEG by representing multi-scale spatio-spectral-temporal features. Section IV presents experimental settings and results by comparing the proposed method and comparable baselines. In Section V, we analyze our proposed method from several points of view. Finally, Section VI summarizes the proposed study and suggests future research directions.

II. Related Work

Learning a class-discriminative feature representation of EEG is still challenging in both theory and practice. Numerous prior studies have attempted to extract features from EEGs. In this section, we briefly discuss linear methods and deep learning models used for EEG signal representation.

A. Linear Models

Over the past decades, CSP [1] and its variants [11], [13] have played an essential role in decoding MI. Blankertz *et al.* [1] and Ang *et al.* [11] independently used a spatial filtering-based method for classifying MI. Ang *et al.* band-pass filtered EEG data before applying CSP, thereby attempting to decode EEG signals in a spatio-spectral manner. They named the

proposed method filter bank CSP (FBCSP). Furthermore, Suk and Lee [13] also decoded MI by jointly optimizing multi-spectral filters in a Bayesian framework and named their method Bayesian spatio-spectral filter optimization (BSSFO). Recently, Jin *et al.* proposed task-related channel selection algorithms by using a correlation-based CSP training [31] and a bispectrum analysis [32]. Jin *et al.* [33] also designed a selection algorithm of CSP-features based on the Dempster-Shafer theory. Since conventional MI decoding systems [1], [11], [13] were mostly developed based on subject-dependent manner that require considerable calibration time, zero-calibration methods have been proposed in recent years [27]. For instance, Lotte, Guan, and Ang [16] exploited multi-resolution spectral decomposition (MR FBCSP) to extract subject-neutral, *i.e.*, subject-independent features. Ray *et al.* [15] also extracted subject-independent features using the FBCSP [11].

CCA is commonly utilized for detecting SSVEP [9] owing to its practical ability to be implemented without the calibration stage. The standard CCA method deployed sinusoidal signals as reference signals and estimated canonical correlation between the reference signals and input EEG signals to identify an evoked frequency in SSVEP EEGs.

In addition, to characterize the sleep stage, entropy calculation-based approaches were frequently used. Sanders *et al.* [35] classified the sleep stage using the spectral-temporal features of EEGs learned from short-time Fourier transformation. Furthermore, Zheng and Lu [36] focused on identifying a driver's mental fatigue during driving. They applied filter banks to EEG signals to extract spectral information, and then transformed the filtered EEG signals to spectral space, *i.e.*, estimated PSD of filtered EEG signals. By doing so, Zheng and Lu effectively assessed the regression score of the driver's mental states which were labeled using the PERCLOS index, a measure of neurophysiological fatigue.

Earlier, Shoen and Guttg [14] applied a machine learning approach to extract and classify the spatio-spectral-temporal features of epileptic seizure EEG signals. Specifically, they used filter banks in a channel-wise manner to capture the spatio-spectral information. Then, by encoding the temporal evolution of extracted spatio-spectral feature vectors, they effectively constructed epileptic seizure EEG signal spatio-spectral-temporal features and classified the seizure and non-seizure features utilizing a support vector machine (SVM). Recently, spectral features derived from a principal component analysis (PCA) exhibited superior performance for seizure onset detection. In particular, Lee *et al.* [37] band-pass filtered raw signals and calculated PSD. Then they applied PCA for the extraction of EEG signal spectral features.

These practical linear model-based BCI methods [1], [9], [11], [13], [35]–[37] have demonstrated credible performance. However, these methods need to have certain prior neurophysiology knowledge [4], because their feature extraction stages are specifically designed for each EEG paradigm. Conversely, our method does not need to be specialized for different paradigms.

B. Deep and Hierarchical Models

Recently, deep learning methods, especially CNNs, have achieved promising results in EEG signal decoding research. For instance, Schirrmeister *et al.* [3] introduced Shallow ConvNet, Deep ConvNet, Hybrid ConvNet, and Residual ConvNet. They evaluated how well various proposed CNNs decoded MI. Ko *et al.* [18] also proposed a novel CNN architecture which is inspired by a recurrent convolutional neural network [38] for MI classification, deep recurrent spatio-temporal neural network (RSTNN). Despite great success in deep learning-based MI decoding methods, some limitations in subject-dependent MI-based BCIs still exist [27]. However, owing to its great benefits such as zero-calibration, some pioneering research [17], [27] proposed deep learning-based subject-independent MI-BCIs. Zhang *et al.* [17] proposed a combined recurrent neural network (RNN) and CNN for extracting spatio-temporal feature representation. Kwon *et al.* [27] exploited multi spectral-spatial features (SSFR) using spectral-spatial filtering and CNN and experimented on the large-scale database.

While a standard CCA [9] has obtained state-of-the-art performance in SSVEP-based BCI, Kwak *et al.* [23] developed a CNN for SSVEP feature representation learning. These authors simply combined spatial and temporal convolution to enable the system to learn data patterns in the latent space, thereby correctly generalizing EEG signal features. Meanwhile, Waytowich *et al.* [22] applied EEGNet [4] to the SSVEP paradigm and achieved a higher performance than that of the standard CCA.

Supratak *et al.* [20] developed a deep neural network for sleep stage detection. More precisely, they combined a CNN for representation learning and a recurrent neural network for sequential residual learning. Furthermore, they trained the deep learning model in two separate steps, optimizing the model by individual pre-training and fine-tuning. In the meantime, Gao *et al.* [26] proposed an EEG-based spatio-temporal convolutional neural network (ESTCNN) for driver fatigue evaluation. The ESTCNN simply convolved the band-pass filtered EEG to represent temporal dependencies and flattened the extracted features for spatial features fusion. Lastly, densely connected layers were used for the identification of a user's condition.

To detect a seizure type, Asif *et al.* [24] proposed a multi-spectral deep feature learning using a deep CNN, SeizureNet. They transformed the EEG signals to spectral space using saliency-encoded spectrogram generation and fed the extracted spectral features to a deep neural network. In the meantime, Emami *et al.* [25] independently proposed another CNN-based approach for detecting seizure onset. They band-pass filtered and segmented the input EEG patterns, and then used a deep CNN for classification.

Recently, Lawhern *et al.* [4], [22] proposed a novel CNN, the so-called EEGNet. Unlike other linear or deep learning-based methods, the EEGNet classified various EEG paradigms using a single architecture, *i.e.*, not specifically tuned for different

paradigms. Furthermore, Lawhern *et al.* used a separable convolution [19] for parameters reduction method.

On one hand, the deep and hierarchical models decoded the EEG signals well without any custom feature extraction stage for their respective paradigms [3], [18], [20], [23]–[26] or even various paradigms [4], [22]. On the other hand, the deep CNNs extracted the EEG features at a sequential level using stacked convolutional layers without exploiting multi-scale spectral representation. Conversely, the proposed method exploits multi-scale spatio-spectral-temporal features irrespective of the input EEG paradigms.

III. Methods

In this section, we propose a deep multi-scale neural network (MSNN), which can represent EEG features from different paradigms by exploiting *spatio-spectral-temporal* information at multi-scale.

A. Multi-Scale Neural Network

As mentioned previously, a FBCSP [11] is one of the most successful models to exploit *multi-scale* EEG signal features, especially for MI. Thus, many successful MI EEG signal decoding algorithms [3], [13] or even other paradigm classification algorithms [4] were inspired by the FBCSP model. In this study, the proposed MSNN also learns multi-scale feature representations. However, the network automatically learns from data through discriminative multiple spectral filters, rather than manually defining multi-frequency bounds as in FBCSP. Basically, our proposed method consists of three types of blocks: (1) a *spectral-temporal* feature representation block, (2) a *spatial* feature representation block, and, (3) a classification block, as depicted in Fig. 2.

First, in the spectral-temporal feature representation block, stacked convolutional layers extract EEG data spectral-temporal features, similar to the existing deep learning-based EEG classification methods [3], [4]. However, the proposed model exploits the intermediate activations for gathering multi-scale spectral information. Then, the spatial feature representation block discovers spatial patterns from the extracted multi-scale features. Finally, these multi-scale spatio-spectral-temporal features are concatenated, pooled, and fed into the densely connected layer for classification.

B. Spectral-Temporal Feature Representation Block

Given an input EEG data $\mathbf{x}$, we reshape it in the form of $[n_c, n_T, 1]$, *i.e.*, $\mathbf{x} \in \mathbb{R}^{n_c \times n_T \times 1}$, where n_c and n_T denote the number of electrode channels and timepoints, respectively.

In the MSNN, the input EEG data is temporally convolved in a channel-wise manner by a spectral convolutional layer to expand the number of feature maps. Thus, the activated features have the form $[n_c, n'_T, F_0]$, where $n'_T = n_T - (f_s/2) + 1$ (f_s and F_0 are the sampling frequency and the feature map dimension for the first temporal convolution layer). The main benefits of using a separable convolution [4], [19] are a significant reduction of tunable weights, in the model and more importantly, an

efficient and explicit decoupling of the relationship between the temporal and the feature map dimensions of the input features. This is accomplished by learning kernels independently for each feature map. Thus, as in BCI literature, the separable convolution [19] enables the system to learn temporal kernels individually from the feature map dimensions (using a depth-wise convolution [19]), and then optimally re-combine the feature maps (using a pointwise convolution [19]).

In this block, by setting a kernel size of $(1 \times T_k)$, where T_k denotes the kernel size of the k -th temporal separable convolution, the k -th temporal separable convolutional layer represents EEG signal features in the range of T_k/f_s' sec, hence f_s'/T_k Hz, where f_s' is a frequency property extracted at the first spectral convolutional layer. Therefore, the spectral-temporal feature representation layers can deal with different timepoints or frequency ranges by using various kernel sizes for the input EEG data.

Additionally, each different layer that has a different kernel size extracts features in different frequency and timepoint ranges. In other words, a spectral-temporal convolution layer with a larger kernel represents *longer-term* temporal features, *i.e.*, a *lower-range* of spectral features and vice versa. Then, the MSNN exploits intermediate activations from each layer, thus learning multi-scale feature representations.

In addition, a separable convolution [19] only operates convolutions in a cross- n_T way, thus, the number of parameters is small compared to a conventional convolution. For instance, while a k -th separable temporal convolution has only $T_k + F_{k-1} \cdot F_k$ parameters, the conventional convolution with the same size kernel has $T_k \cdot F_{k-1} \cdot F_k$ parameters, where F_k denotes the feature maps dimension of the k -th layer.

Furthermore, in this processing, as described above, the MSNN uses its intermediate activations to exploit multi-scale representations. In other words, the proposed network obtains N numbers of spectral-temporal features $\mathbf{f}_k^{\text{ST}}, k = 1, 2, \dots, N$ like:

$$\mathbf{f}_k^{\text{ST}} = \mathbf{C}_k^{\text{ST}}(\mathbf{x}) = \mathbf{C}_k^{\text{sep}} \circ \mathbf{C}_{k-1}^{\text{sep}} \circ \dots \circ \mathbf{C}_0(\mathbf{x}), \quad (1)$$

where $\mathbf{C}_k^{\text{sep}}, \mathbf{C}_0$, and $\mathcal{F}_i \circ \mathcal{F}_j$, respectively, denote the k -th separable convolution, the first temporal convolution, and a function composition between arbitrary functions $\mathcal{F}_i$ and $\mathcal{F}_j$, *i.e.*, $\mathcal{F}_i \circ \mathcal{F}_j(\cdot) = \mathcal{F}_i(\mathcal{F}_j(\cdot))$. Thus, by extracting features $\mathbf{f}_1^{\text{ST}}, \mathbf{f}_2^{\text{ST}}, \dots, \mathbf{f}_N^{\text{ST}}$, the MSNN effectively represents the spectral-temporal features from the multi-scale viewpoint, thereby automatically enhancing generalization. In addition, as all inputs are *zero-padded* before each separable temporal convolution, the output features have the same dimension for the channels and timepoints, except for the feature map dimension. Thus, the k -th spectral-temporal feature $\mathbf{f}_k^{\text{ST}}$ now has the form $[n_c, n'_T, F_k]$.

In summary, this spectral-temporal representation block in our work extract spectral features of EEG signals at different band-pass frequencies by convolving them with $(1 \times (f_s/2))$ kernel. Specifically, based on Nyquist-Shannon sampling theory, setting the kernel as $(1 \times (f_s/2))$ allows our model to

capture spectral information at 2 Hz and above by multi-dimensional output filters, *i.e.*, F_0 , [4], [22]. Then, the following temporal separable convolutions with different kernel sizes $(1 \times T_k), \forall k = 1, \dots, N$, convolve different spectral-temporal features in a filter-independent manner. A brief graphical overview of this spectral-temporal feature representation block is illustrated in Fig. 3(a).

C. Spatial Feature Representation Block

In the spatial feature representation block, a common spatial convolution is used for feature extraction. In this block, the kernel size is constrained to be equal to the number of EEG channels; hence, a convolution with a kernel of $(n_c \times 1)$ is used. Additionally, by setting the kernel size to be the same as the

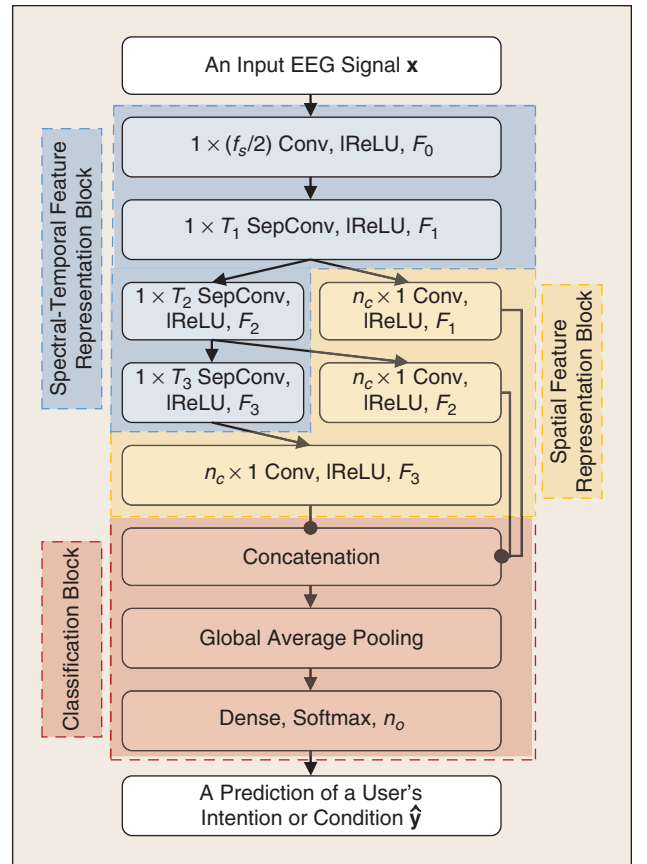


FIGURE 2 The proposed architectural framework of our multi-scale neural network (MSNN). In the proposed network, first, an input EEG $\mathbf{x}$ is temporally convolved (Conv) to expand the number of features, where f_s, F_0 , and IReLU denote the sampling frequency rate, the number of output filter maps of the first layer, and a leaky rectified linear unit activation function respectively. Then, a set of temporal separable convolutions (SepConv) extracts *spectral-temporal* features (T_k and F_k denote, respectively, the kernel size and output feature maps of k -th temporal separable convolution, $k = 1, 2, \dots, N$, where N is the number of temporal convolution layers). At the same time, a set of spatial convolutions represents *spatial* features, where n_c denotes the number of acquired EEG channels. Then, the *multi-scale* features are concatenated and fed into the global average pooling layer [39]. Finally, the dense layer (Dense) determines the class of an input EEG by exploiting multi-scale features, where n_o denotes the number of output nodes.

number of electrode channels, similar to many existing deep learning-based BCI methods [3], [4], [18], the proposed MSNN extracts spatial information from the original EEG acquisition channel distributions of multi-scale spectral-temporal features. Then, the MSNN can obtain *neurophysiologically plausible* information from the input data distribution.

Furthermore, the spatial feature representation can be applied unrestrictedly, thus in the proposed method, we add this block after every extracted spectral-temporal feature $\mathbf{f}_k^{\text{ST}}, k = 1, 2, \dots, N$ like,

$$\mathbf{f}_k^{\text{SST}} = \mathcal{S}_k(\mathbf{f}_k^{\text{ST}}) = \mathcal{S}_k \circ \mathcal{C}_k^{\text{ST}}(\mathbf{x}), \quad (2)$$

where $\mathcal{S}_k$ denotes the k -th spatial convolution and $\mathbf{f}_k^{\text{SST}}$ are *spatio-spectral-temporal* features estimated by the $\mathcal{S}_k$ and $\mathcal{C}_k^{\text{ST}}$. We use *valid paddings* for every spatial convolution, thus the k -th spatio-spectral-temporal feature $\mathbf{f}_k^{\text{SST}}$ has the form $[1, n'_T, F_k]$. By setting the number of spatial convolutions to be identical to the number of spectral-temporal convolutions, unlike many previous research using deep learning for BCI [3], [4], [17], [18], [20], [24], we extract spatial features of each range from spectral-temporal features. In other words, unlike many previous stacked CNNs, the proposed architecture uses every intermediate activated feature set to exploit spatial information, thereby creating the capability to extract various ranges of EEG features at multi-scale. Refer to Fig. 3(b) for the graphical overview of the spatial feature representation block.

D. Classification Block

For classifier learning, because we have N numbers of different (or same when $F_1 = F_2 = \dots = F_N$) size of spatio-spectral-temporal features $\mathbf{f}_k^{\text{SST}}, k = 1, 2, \dots, N$, the classifier in the proposed method has to concatenate the features in the feature map dimension. Thus, the concatenated feature $\mathbf{f}_{\text{concat}}^{\text{SST}}$ is represented as:

$$\mathbf{f}_{\text{concat}}^{\text{SST}} = \parallel_{i=1}^N \mathbf{f}_i^{\text{SST}} = \parallel_{i=1}^N \mathcal{S}_i \circ \mathcal{C}_i^{\text{ST}}(\mathbf{x}), \quad (3)$$

where $\parallel$ denotes the concatenation operation.

For the classifier network, let us assume that the number of output classification nodes is denoted by n_o and we use a single linear mapping layer. Then, we need to train the large number of $\sum_{i=1}^N n_o \cdot n'_T \cdot F_i$ parameters (note that we disregard the bias term for simplicity) because $\mathbf{f}_{\text{concat}}^{\text{SST}}$ has the form $[1, n'_T, \sum_{i=1}^N F_i]$, and it would still require a large number of training samples. Therefore, after representing the input EEG data to multi-scale spatio-spectral-temporal features, the proposed MSNN has one extra operation for reducing the trainable weights. Unlike the existing deep learning-based BCI methods [3], [4], [17], [18], [20], [23], [24], global average pooling (GAP), which is widely used in the computer vision field [39], is performed.

The GAP layer [39], a type of pooling layer, averages nodes from each feature map, thus eliminating the requirement for any window size or stride. By applying GAP, our proposed MSNN efficiently extracts significant features. From the BCI literature, the GAP layer can be understood to be a method that can emphasize an important frequency range and its surrounding area for each feature map dimension. Thus, for the extracted multi-scale features in the MSNN, the GAP layer [39] stresses the crucial spectral-temporal part resulting in concise information for the final decision making.

Additionally, the GAP layer [39] significantly reduces the number of classifier parameters used in the proposed MSNN. Specifically, after the GAP layer $\mathcal{G}(\cdot)$, the extracted feature is reduced to the form $[1, 1, \sum_{i=1}^N F_i]$, whereas the feature without GAP has the form $[1, n'_T, \sum_{i=1}^N F_i]$. Therefore, we drastically suppress the trainable parameters in the classifier from $n'_T \cdot n_o \cdot \sum_{i=1}^N F_i$ to $n_o \cdot \sum_{i=1}^N F_i$.

Then, the MSNN prediction, $\hat{\mathbf{y}}$, for the input EEG data, $\mathbf{x}$, is as follows:

$$\begin{aligned} \hat{\mathbf{y}} &= \text{softmax}(\mathbf{W}_o^T \cdot \mathcal{G}(\mathbf{f}_{\text{concat}}^{\text{SST}}) + \mathbf{b}_o) \\ &= \text{softmax}\left(\mathbf{W}_o^T \cdot \mathcal{G}\left(\parallel_{i=1}^N \mathcal{S}_i \circ \mathcal{C}_i^{\text{ST}}(\mathbf{x})\right) + \mathbf{b}_o\right), \end{aligned} \quad (4)$$

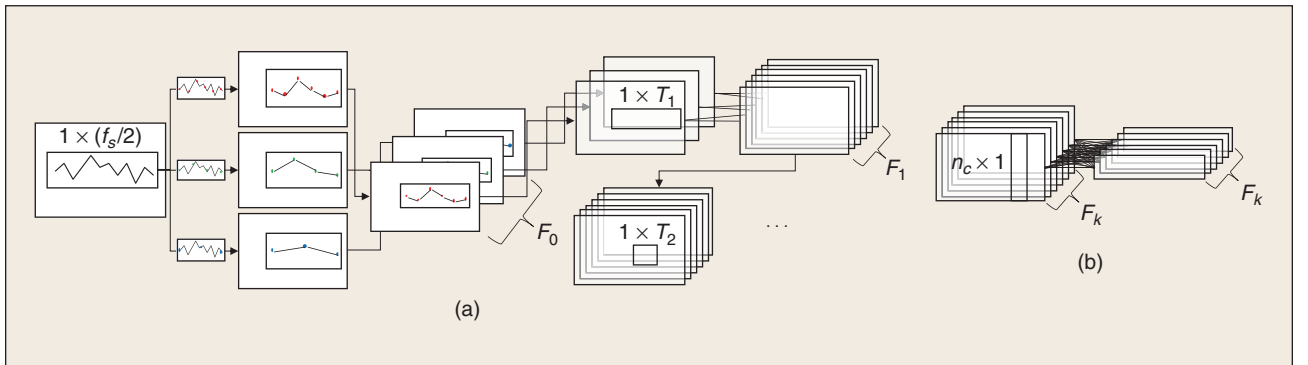


FIGURE 3 A graphical overview of a spectral-temporal feature representation block and a spatial feature representation block. (a) A spectral-temporal feature representation block. The $(1 \times (f_s/2))$ convolution captures spectral information at 2Hz and above by multi-dimensional output filters. Then, the following separable temporal convolutions convolve different spectral-temporal features in a filter-independent manner. (b) A spatial feature representation block. The $(n_c \times 1)$ convolution obtains neurophysiologically plausible spatial information from the input data distribution.

where $\mathbf{W}_o \in \mathbb{R}^{\sum_{i=1}^N F_i \times n_o}$ and $\mathbf{b}_o \in \mathbb{R}^{n_o}$ respectively denote the weight matrix and bias of the classifier.

Finally, the cross-entropy loss, $\mathcal{L}$, that is used for network training, is calculated by the prediction $\hat{\mathbf{y}}$ and the label $\mathbf{y}$:

$$\mathcal{L} = \text{CE}(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{b=1}^B \mathbf{y}^{(b)} \log \hat{\mathbf{y}}^{(b)}, \quad (5)$$

where B and $\text{CE}(\cdot, \cdot)$ respectively denote the mini-batch sizes and the cross-entropy loss function, and $\hat{\mathbf{y}}^{(b)}$ and $\mathbf{y}^{(b)}$ denote the prediction and ground-truth label for the b -th training sample in the mini-batch¹.

IV. Experiments

In this section, we describe the datasets used for performance evaluation, our experimental settings, and baseline settings. Furthermore, we present the performance of our method and competing methods.

A. Datasets and Preprocessing

In this study, we used five different publicly available datasets to validate the proposed method on four different EEG data paradigms. Table I shows a brief summary of the datasets used in this work.

1) Motor Imagery (subject-dependent)

First, we used two big datasets for MI EEGs, GIST-MI [40]² and KU-MI [8]^{3,4}. The GIST-MI dataset consists of two different MI tasks: left-hand and right-hand MI that are acquired from 52 subjects. All EEG signals were recorded from 64 Ag/AgCl electrode channels according to the standard 10–20 system, sampled at 512 Hz. Each class contained 100 or 120 trials, and each trial was a 3 sec long MI task. Since this dataset is not separated into training and test samples, we conducted a five-fold cross-validation for a fair evaluation. Additionally, the KU-MI dataset is acquired across two sessions from 54 subjects, recorded from 62 Ag/AgCl electrodes according to the stan-

dard 10–20 system, and sampled with 1000 Hz. Each class of this dataset contains 50 trials with 4 sec long of MI. The KU-MI dataset is originally separated by the training and testing samples, thus we used it directly as training and testing sets. For both MI datasets, we preprocessed signals by applying a large Laplacian filtering,⁵ baseline correction by subtracting the mean value of the fixation signal from each MI trial, and band-pass filtering between 4 and 40 Hz with 4th order Butterworth filter. Then, we removed parts in the first and last 0.5 sec from each trial, and finally applied Gaussian normalization. We applied the same mean and standard deviation values for normalization to the test samples. The multi-channel EEG signals were only shifted and scaled by their respective channel-wise mean and standard deviation values. Thus, inter-channel correlations inherent in the data were preserved.

2) Motor Imagery (subject-independent)

In this work, we mostly focused on developing a subject-dependent BCI method. However, since spontaneous EEG, *e.g.*, MI signals, from different subjects are highly variable [27], [29], developing a deep learning-based subject-independent MI-based BCI method has gained lots of interest recently by pioneering research [17], [27].

In this respect, we conducted extra experiments in the subject-independent manner over KU-MI dataset [8]. In detail, by following [27], we used a *leave-one-subject-out* cross-validation for the performance evaluation. We trained our proposed MSNN using all source subjects' samples and evaluated on the target subject's test samples in session 2.

3) Steady-State Visually Evoked Potentials

We also used the KU-SSVEP dataset [8]³ for SSVEP decoding experiments in this study. This KU-SSVEP dataset was acquired from 54 subjects and recorded from 62 Ag/AgCl electrode channels using the 10–20 system. The KU-SSVEP dataset contains four EEG classes with 4 sec long from target stimuli at 5.45, 6.67, 8.57, and 12 Hz, and each class has 25 EEG trials of training and testing samples for each session. As it is reported that the length of used data has a significant influence on the performance of the SSVEP detection, we have also

¹All codes used in our experiments are available at 'https://github.com/DeepBCI/Deep-BCI/tree/master/1_Intelligent_BCI/Multi_Scale_Neural_Network_for_EEG_Representation_Learning_in_BCI.'

²Available at <http://gigadb.org/dataset/100295>

³Available at <http://gigadb.org/dataset/100542>

⁴Experimental results of the KU-MI dataset are reported in Supplementary B.

⁵When the target channel does not have four nearest neighbors, we just used available channels and their average value to filter the target channel.

TABLE I A brief summary of datasets used in our work. Note that CHB-MIT [12] dataset has different numbers of records for each subject, thus the mean of records are reported at '# sessions'. For fair comparison, we evaluated GIST-MI [40] and SEED-VIG [36] by 5-fold cross-validation (CV) and CHB-MIT [12] by leave-one-record-out (LORO)-CV. Finally, KU-MI and KU-SSVEP [8] datasets are provided with disjoint training/testing samples, so we used them directly.

DATASET	# SUBJECTS	# SESSIONS	# TOTAL TRIALS	# ELECTRODES	SAMPLING RATE [HZ]	TRIAL TIME [SEC]	EVALUATION STRATEGY
GIST-MI [40]	52	1	100 or 120	64	512	3	5-FOLD CV
KU-MI & SSVEP [8]	54	2	200	62	1000	4	TRAIN/TEST
SEED-VIG [36]	1	23	885	17	200	8	5-FOLD CV
CHB-MIT [12]	24	26.58	—	23	256	—	LORO-CV

conducted the experiments with shorter length data.⁶ We pre-processed the SSVEP signals by applying band-pass filtering between 4 and 15 Hz and selected eight channels in the occipital region, 'PO3, POz, PO4, PO9, O1, Oz, O2, and PO10,' because this region is widely used for SSVEP classification [22].

4) Drowsiness

With respect to passive BCI [6], we considered two different paradigms, seizure EEG signals [12] and vigilance EEG signals [36]. Owing to its theoretical and practical benefits, in this study, we conducted experiments identifying drivers' mental fatigue. We also used a publicly available SEED-VIG EEG dataset [36]⁷ for the drowsy driving task data. This dataset consists of 23 experiments, *i.e.*, trials, and each trial is recorded for approximately 2 hours while simulated driving occurs. The EEG signals are acquired from 17 electrode channels according to the 10–20 system and sampled at 200 Hz. For this dataset, we band-pass filtered EEG signals in the range between 0.5 and 40 Hz, each epoch was 8 sec in length. Since the dataset was originally labeled using *PERCLOS* levels [36], we categorized the label vectors into three classes, *awake*, *tired*, and *drowsy*, with two threshold values (0.35 and 0.7). Then, for the 23 experiments, a five-fold cross-validation was used for performance estimations.

5) Seizure

Finally, we conducted seizure onset detection experiments with the widely used and publicly available CHB-MIT dataset [12]⁸. The CHB-MIT dataset contains EEG data from 24 subjects sampled at 256 Hz acquired from 23 electrode channels (24 or 26 in a few cases) according to the 10–20 system. In this work, we selected EEG trials that have the same 23 channels montage and removed some trials acquired from the different montage. By following [14], we used a *leave-one-record-out* cross-validation. More precisely, we trained the proposed method using all non-seizure records and all seizure records but one, and tested the model on the remaining seizure record. Then, we repeated this process for the number of seizure records in the dataset, thus, each seizure record was tested. For training, the test trial epochs were 10 sec in length. During validation and testing sessions, a 10 sec length EEG signal was fed into our proposed network using a 1/256 sliding. In other words, we used epochs in units of 10 seconds in the training session, but a sliding-window of 10 sec length in the testing session. Then, we observed how the probability of ictal or normal for each EEG signal timepoint was continuously changing over time.

For all datasets, we exploited *early stopping* strategy. More specifically, we randomly selected and split again training samples at a 9:1 ratio for each experiment and used them for training and model selection respectively.

B. Experimental Settings

In our work, we compared our method with paradigm-specific linear model-based and deep learning-based methods for each EEG paradigm.

1) Linear Models–Motor Imagery

First, we built a CSP with a linear discriminant analysis (CSP + LDA) [1] and an FBCSP with an LDA (FBCSP + LDA) [11] for MI decoding. We used four filters and regularized covariance for the CSP and FBCSP. Additionally, we also used nine non-overlapped filter banks in the 4 ~ 40 Hz range, *i.e.*, 4 ~ 8, 8 ~ 12, ..., 36 ~ 40 Hz, and finally selected 10 features using the mutual information-based feature selection method FBCSP. Finally, for the BSSFO [13], we set 30 particles on the 4 ~ 40 Hz range MI-EEG to achieve robust classification results and classified the extracted features with a SVM (BSSFO + SVM).

2) Linear Models–Steady-State Visually Evoked Potentials

We also built a standard CCA [9] for SSVEP classification. We set reference signals for each stimulus including second harmonics. Furthermore, the standard CCA does not require training samples for the optimization, thus we only estimated each session in its entirety from the KU-SSVEP dataset [8] for the CCA performance estimation.

3) Linear Models–Drowsiness

For the drowsy state detection experiment, we estimated the filter-banked input EEG data PSD in a channel-wise manner for extracting spatio-spectral features and classified the learned features using an SVM with a radial basis function (RBF) kernel ($\gamma = 1/d_{\text{input}}$ where d_{input} denotes the input feature dimension) [36].

4) Linear Models–Seizure

In addition, we also reimplemented Shoeb and Gutttag [14]'s method for the seizure onset detection experiment. We applied the PSD to the EEG data in a channel-wise manner. Then, the 3 sec time window time evolution method was used for capturing temporal information. Finally, the represented spatio-spectral-temporal features were fed into an SVM using an RBF kernel ($\gamma = 1/d_{\text{input}}$).

5) Deep Neural Networks–Motor Imagery

We also implemented deep learning-based BCI models⁹ for MI. Basically, most of the existing deep learning models [3], [17], [18], [25], [26] have focused on a paradigm-specific BCI task. However, we conducted experiments over all types of datasets for each deep learning model to demonstrate the validity of the proposed method. We built a Shallow ConvNet and a Deep ConvNet as proposed by Schirrmeister *et al.* [3]. The Shallow ConvNet consists of two convolutions, temporal and

⁶Experimental results of the KU-SSVEP dataset with shorter length are reported in Supplementary B.

⁷Available at: <http://bcmi.sjtu.edu.cn/seed/download.html>

⁸Available at: <https://physionet.org/content/chbmit/1.0.0/>

⁹See 'Supplementary A: Architectural Details of Deep Models for BCIs' for more detail architectures and learning schedules.

spatial, with a squaring nonlinear activation, an average pooling, and a logarithmic activation. The Deep ConvNet [3] has five convolutions, temporal and spatial, and three additional temporal convolutions. The RSTNN [18] is also used for these experiments. This network consists of three recurrent convolutional layers, and each recurrent convolutional layer has three recurrent temporal convolutions [38] with a spatial convolution. Additionally, we also conducted experiments on two combined deep neural networks, ‘parallel’ and ‘cascade’ convolutional recurrent networks (Parallel CRN and Cascade CRN) proposed by Zhang *et al.* [17]. In both models, EEG signal vectors were converted to 2D meshed data according to the acquired electrode map. In the parallel structure, a three-layer CNN and a two-layer RNN simultaneously extracted spatial and temporal features respectively, and a concatenation is used for the feature decomposition. Furthermore, in the cascade structure, a three-layer CNN extracts spatial features of mesh data at each timepoint, and the extracted features are fed into a two-layer RNN to represent temporal information. Note that we only conducted experiments on the MI and SSVEP datasets [8], [40] due to the lack of electrode channels of other paradigms to generate 2D meshed data.

6) Deep Neural Networks–Steady-State Visually Evoked Potentials

For the SSVEP decoding experiment, we exploited another version of EEGNet for SSVEP EEG [22]. We used different kernel sizes for this EEGNet as Waytowich *et al.* proposed. The SSVEP classification performance estimated by this version is marked by † in the classification table.

7) Deep Neural Networks–Drowsiness

The ESTCNN [26] which is proposed for mental fatigue classification has three core blocks. Each block in the ESTCNN consists of three temporal convolutions with a max pooling layer, with the exception of the last block that uses an average pooling layer instead of the max pooling.

8) Deep Neural Networks–Multi-paradigm

Finally, we also implemented the EEGNet [4] in our study. As previously mentioned, we used different kernel sizes for two different EEGNets, [4] and [22]. Nevertheless, the basic architecture of the network was the same for various EEG paradigms, having a temporal convolution, depthwise spatial convolution [19], and separable temporal convolution [19].

9) Proposed Multi-Scale Neural Network

While training our proposed network, depicted in Fig. 2, we set a mini-batch size of 16, an exponentially decreasing learning rate (initial value: 0.03, decreasing ratio per epoch: 0.001), and an Adam optimizer. For the first temporal convolution, we used a conventional temporal convolution with the kernel size of $(1 \times f_s/2)$ and $F_0 = 4$. Furthermore, we used three spectral-temporal feature representation convolutions, *i.e.*, $N = 3$, and set $T_1 = 100$, $T_2 = 60$, and $T_3 = 20$ with $F_1 = 16$, $F_2 = 32$,

and $F_3 = 64$. Then, for the spatial feature representation block, we used three spatial convolutions because the number of spatial convolutional layers must be the same as the number of spectral-temporal separable convolutional layers. The proposed method used different kernel sizes for the SSVEP dataset, similar to the EEGNet [22] due to the fact that SSVEP EEG data is created by target frequencies [8], [9]. For the KU-SSVEP dataset [8], we set $T_1 = 20$, $T_2 = 10$, and $T_3 = 5$ for the spectral-temporal feature representation block, and used the same settings for the others. The SSVEP classification performance estimated by this method is marked by ‡. Additionally, batch normalization was performed after every convolution. Finally, for the classification block, all activated features from the *spatio-spectral-temporal* block were concatenated and fed into the GAP [39] layer. Then, after flattening, the *multi-scale* features were linearly mapped by a dense layer. In this proposed network, a leaky rectified linear unit (ReLU) activation function, a L1-L2 regularizer ($\ell_1 = 0.01$ and $\ell_2 = 0.001$), and a Xavier initializer [41] are used for all tunable parameters except for the final decision layer that is activated by a softmax activation function instead of a leaky ReLU. We selected model components that demonstrated the best performance for validation, *i.e.*, model selection samples, as mentioned previously.

C. Experimental Results

In this section, we summarized all experimental results in Table II. Furthermore, we performed two-tailed Wilcoxon’s signed-rank test to estimate *p*-values of competing baseline models and our proposed MSNN, comparing performance differences statistically.

1) Motor Imagery (subject-dependent)

Our proposed network clearly outperformed other baselines for MI EEG signal decoding. Importantly, the proposed network achieved a higher accuracy than the methods designed specifically for MI classification: CSP [1], FBCSP [11], Shallow ConvNet [3], Deep ConvNet [3], and RSTNN [18]. With this clear improvement in accuracy, we could expect that our proposed method is one step closer to MI-based BCI commercialization.

2) Motor Imagery (subject-independent)

By experimental results summarized in Table III, we conclude that our proposed MSNN is *well-generalized* not only in the subject-dependent manner, but also in the subject-independent manner. Based on this promising result, we also conclude that our method can be used for the zero-calibration in MI-based BCIs.

3) Steady-State Visually Evoked Potentials

Our proposed MSNN achieved a slightly lower performance than CCA [9], Deep ConvNet [3], and EEGNet [22] in the SSVEP classification. However, the difference in performance between our MSNN and the other three baselines, CCA [9], Deep ConvNet [3], and EEGNet [22], was reasonably small and the proposed method performed with a credible accuracy score.

4) Drowsiness

In our experiments, we did not estimate accuracy but the number of false detections for the drowsiness classification because the drowsiness dataset has unbalanced numbers for each class. The proposed MSNN made the smallest number of mistakes in decision making for passive BCI [6]. In particular, the proposed method detected a driver's mental fatigue, *i.e.*, drowsiness, from the EEG signals. Our proposed method predicted 31.10 incorrect trials from a total of 177 samples on average. Furthermore, accurately detecting a drowsy state is one of the most important MSNN capabilities for practical use. Our proposed model only made 5.38 mistakes out of 35 drowsy trials on average, thus exhibiting the highest precision score.

5) Seizure

Finally, for the seizure detection task, we also estimated the number of false detections by following [14]. The MSNN incorrectly identified 5.35 seizures among 178 total test seizure samples. Furthermore, our proposed network was the fastest for

detecting seizures, *i.e.*, it exhibited the shortest latency time¹⁰ (approximately 4.98 sec on average) among various methods. In other words, our proposed method demonstrated the best performance even with the shortest latency time. Additionally, the proposed model correctly identified approximately 92% of the seizures within 4.98 sec. We do not present the standard deviation values for this seizure detection experiment because each test trial consisted of different numbers of seizures.¹¹

V. Analyses and Discussions

In this section, we analyzed our proposed network. We determined the feature response by estimating PSD values and relevance scores [42] to show the multi-scale learning benefits. We also visualized learned weights and represented features of the

¹⁰To the best of our knowledge, it is still challenging to detect seizure onset early in time (*i.e.*, before the onset or short latency after the onset) [12], [14], [37].

¹¹See 'Supplementary B: Additional Experiment' for more performance comparison with other methods.

TABLE II Performance evaluations. The Method column denotes all used classification/detection methods including baselines and the proposed method on the various datasets, GIST-MI [40], KU-SSVEP [8], SEED-VIG [36], and CHB-MIT [12] EEG dataset. Each cell depicts the average performance and the standard deviation of all subjects (or trials for the SEED-VIG [36]). For classification performance on the SSVEP dataset, we used different kernel sizes for EEGNet [22] and the proposed method. These values are marked by † and ‡, respectively. Furthermore, we estimated the number of false detections for SEED-VIG [36] and CHB-MIT [12] dataset in the practical standpoint. * and ** respectively denote $p < 0.05$ and $p < 0.01$ estimated by two-tailed Wilcoxon's signed-rank test between each Method and the proposed MSNN.

METHOD	GIST-MI [40]		KU-SSVEP [8]		SEED-VIG [36]		CHB-MIT [12]
	CLASSIFICATION ACCURACY				NUMBER OF FALSE DETECTIONS		
	MEAN±STD.				MEAN±STD.	FALSE POSITIVE (DROWSY)	MEAN (MEAN LATENCY)
CSP + LDA [1]	.66±.14**	—	—	—	—	—	—
FBCSP + LDA [11]	.68±.15**	—	—	—	—	—	—
BSSFO + SVM [13]	.72±.17**	—	—	—	—	—	—
CCA [9]	—	.94±.09*	—	—	—	—	—
PSD + SVM [36]	—	—	31.20±15.47	—	6.74	—	—
SHOEB AND GUTTAG [14]	—	—	—	—	—	—	5.35 (5.11)
SHALLOW CONVNET [3]	.63±.11**	.52±.20**	34.89±19.13	—	6.51	—	19.21** (8.48)
DEEP CONVNET [3]	.61±.07**	.96±.08**	41.31±21.04**	—	8.65	—	8.74** (7.52)
RSTNN [18]	.69±.12**	.65±.20**	39.84±22.56**	—	8.08	—	24.35** (9.31)
ESTCNN [26]	.67±.10**	.79±.17**	41.10±21.31**	—	8.71	—	6.41** (7.01)
EEGNET [4], [22]	.64±.07**	.93±.10†	46.63±22.10**	—	11.26	—	5.40** (6.23)
PARALLEL CRN [17]	.79±.11**	.93±.10	40.51±20.08**	—	9.32	—	12.45** (8.41)
CASCADE CRN [17]	.72±.09**	.94±.08	42.24±19.84**	—	9.84	—	18.37** (7.54)
MSNN (PROPOSED)	.81±.12	.93±.08‡	31.10±17.29	—	5.38	—	5.35 (4.98)

TABLE III Performance evaluations on the subject-independent manner. The Method row denotes all used classification methods including baselines and the proposed method on the KU-MI [8] dataset. Each cell depicts the average performance and the standard deviation of all target subjects. * denotes evaluation performances taken from [27].

METHOD	POOLED CSP [16]	RAY ET AL. [15]	MR FBCSP [16]	PARALLEL CRN [17]	CASCADE CRN [17]	SSFR [27]	MSNN (PROPOSED)
ACCURACY	.66±.16*	.67±.16*	.69±.15*	.72±.16	.73±.15	.74±.16*	.74±.18

proposed method using different methodologies, activation pattern maps [30] and t-SNE plots. Additionally, we observed a practical use for the proposed method, especially for drowsiness and seizure detection experiments.

A. Multi-Scale EEG Feature Extraction

To demonstrate the multi-scale information capture ability of our proposed method, we estimated and plotted PSD values and relevance scores for MI EEG samples. Specifically, we estimated PSD values for subject 48 and 52 in the GIST-MI dataset [40]'s EEG samples from channels on the motor cortex. Additionally, we calculated relevance scores for those subjects using a *layer-wise relevance propagation* [42]. In our results, all classification methods evenly demonstrated good generalization (baselines: $\sim 85\%$ and proposed: $\sim 90\%$) for subject 48, whereas only the proposed method achieved superior performance for subject 52 (baselines: $< 65\%$ and proposed: $\sim 80\%$). As Fig. 4 shows, subject 48's EEG samples are highly activated at the μ range, while subject 52's samples do not show any clear trend at the μ range, but in a wider range. In the meantime, in our class-relevance score visualization (Fig. 4), we could observe the relation between the PSD characteristics of the input EEG and the features' relevance to the output class decision. That is, for subject 48 (Fig. 4(a)), features of $\mathcal{G}(\mathbf{f}_1^{\text{SST}})$ showed higher class-relevance scores than those of other scales $\mathcal{G}(\mathbf{f}_2^{\text{SST}})$ and $\mathcal{G}(\mathbf{f}_3^{\text{SST}})$. Meanwhile, for subject 52 (Fig. 4(b)), features of all scales, *i.e.*, $\mathcal{G}(\mathbf{f}_1^{\text{SST}})$, $\mathcal{G}(\mathbf{f}_2^{\text{SST}})$, and $\mathcal{G}(\mathbf{f}_3^{\text{SST}})$ showed similar rele-

vance scores, meaning that most of those features were involved in making the class decision.

From this phenomenon, we can conclude that our proposed MSNN can capture important features on the multi-scale range, not only in the frequency of interest. In other words, while other existing methods gather spatio-spectral-temporal information at the sequential level, the proposed network exploits multi-scale features, thereby improving learning ability.¹²

B. Activation Patterns

Earlier, Haufe *et al.* [30] proposed an *activation pattern* which is based on a *forward-backward modeling* in signal processing. The activation pattern method [30] provides a way to interpret weight matrices in multivariate neuroimaging, as presented in the signal processing literature.

The proposed method, clearly, decodes the input EEG signal to the corresponding label, *i.e.*, inferring a user's intention or condition from an observed EEG pattern. Therefore, it is a backward process computational model. Hence, for a concrete and meaningful understanding of learned layers, it is essential to reverse this backward process model to a forward process. Finally, in this work, we estimated and visualized the activation patterns of the learned weights shown in Fig. 5(a). We extracted the spatial convolutions of Shallow ConvNet [3], Deep ConvNet [3], RSTNN [18], EEGNet [4], and the proposed

¹²Randomly selected additional results are reported in Supplementary C.

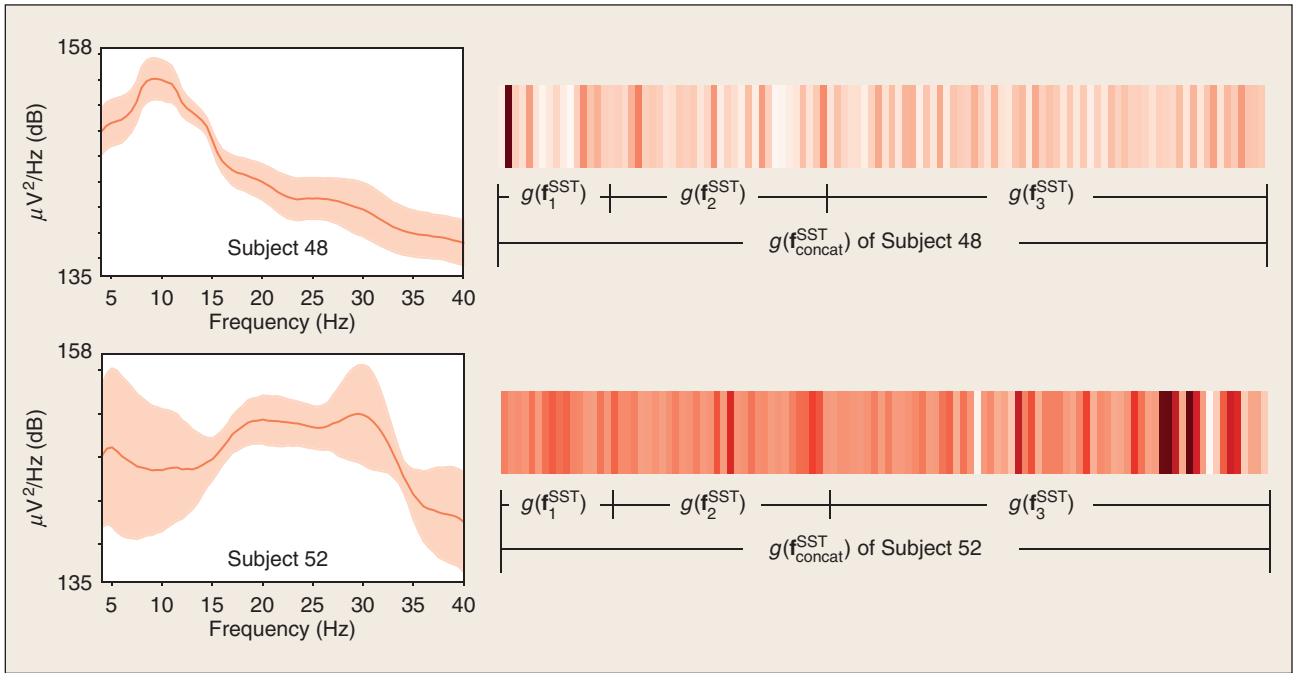


FIGURE 4 PSD curves (left) and relevance scores [42] (right) for subject 48 (top) and subject 52 (bottom) from the GIST-MI dataset [40]. For the PSD curves, the solid red line and the shaded region exhibit the mean and standard deviation of PSD values of all trials, respectively. For the relevance scores, $\mathcal{G}(\mathbf{f}_k^{\text{SST}})$ denotes the global average pooled features of the k th spectro-temporal layer and the deeper red color denotes a higher relevance score. We observed that our proposed MSNN concentrates features from the lower frequency range for subject 48 and a wider range for subject 52 from the relevance scores. This observation supports that our proposed MSNN is able to represent neurophysiological patterns from the input EEG signals.

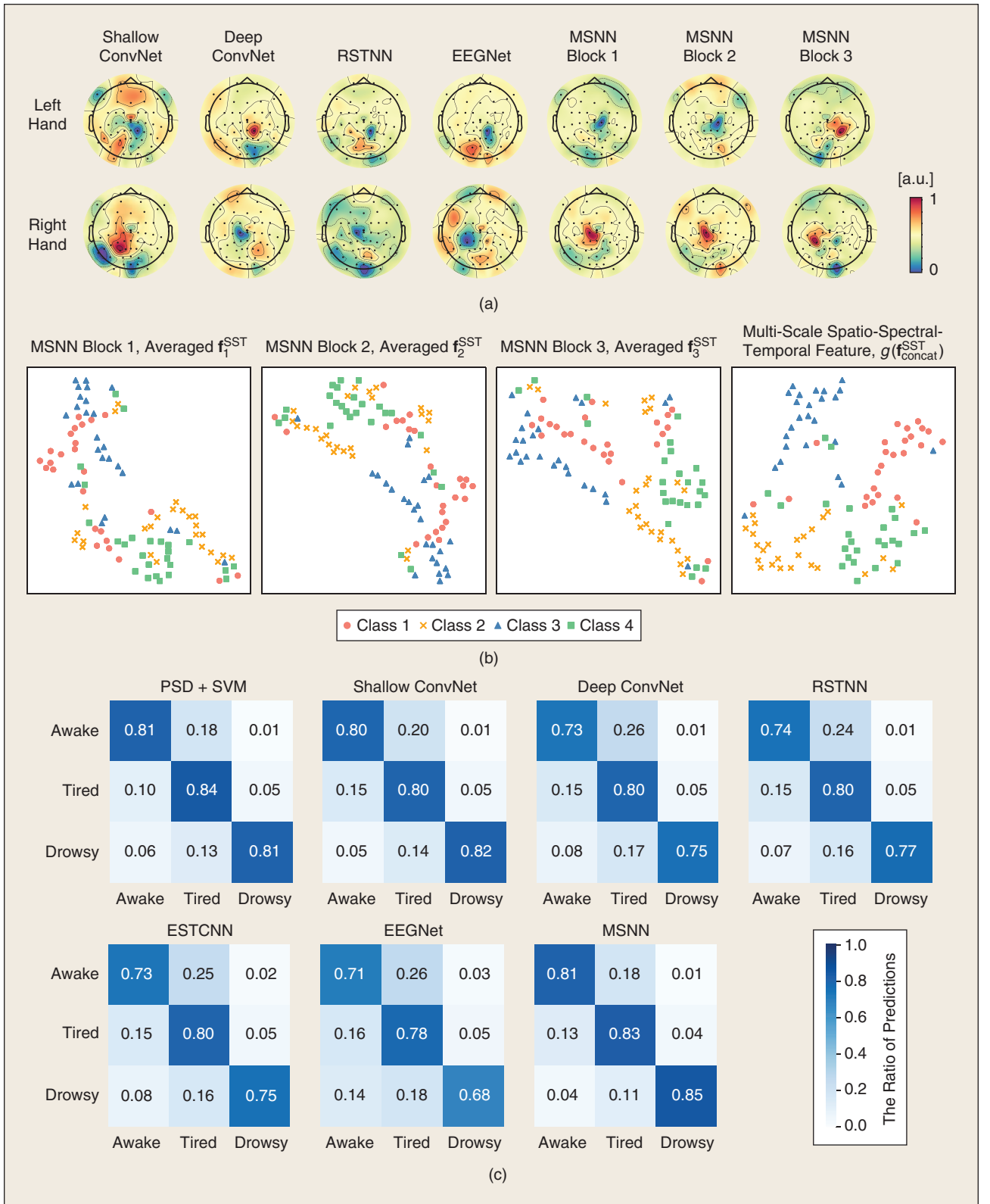


FIGURE 5 Investigation of learned weights (Fig. 5(a)) and represented features (Fig. 5(b)), and inspection of the practical usage of the proposed network (Fig. 5(c) and 5(d)). (a) Topologically visualized activation pattern maps [30] of comparable baselines, and three spatial convolutions in the proposed network. All these visualized patterns here are estimated by the first subject's first fold EEG signals in the GIST-MI dataset [40] and normalized in a range between 0 and 1. Finally, [a.u.] denotes an arbitrary unit. Note that ESTCNN [26], Parallel CRN [17], and Cascade CRN have no spatial convolutions, thus their patterns are not visualized. (b) Visualization of t-SNE transformed represented features for test SSVEP EEG samples. The first three figures denote extracted features by the first, second, and final spatial convolutional layers of the proposed method. The final figure exhibits the GAP [39]-ed feature, $g(f_{\text{concat}}^{\text{SST}})$ which is used for final decision making. (c) Normalized averaged confusion matrices estimated by comparable baselines and the proposed method using the SEED-VIG dataset [36]. (Continued)

model. Then, we estimated activation patterns and visualized them in a topological manner. We do not estimate ESTCNN [26] activation patterns because the ESTCNN [26] does not have any spatial feature representation layers and the visualized patterns are estimated by the first subject's first fold data in the GIST-MI dataset [40]. Finally, we normalized the activation patterns in $[0, 1]$ range before visualization.

In this investigation, we observed right-lateralized brain activation/deactivation patterns, and the same patterns in the left hemisphere when a user imagined the movement of the left hand and right hand respectively. Since there is no golden rule to measure a discriminability, we exploited a cosine similarity score.¹³ Based on our estimation, our proposed method achieved the smallest cosine similarity (0.0047) among the other baselines (0.0224 ~ 0.2750), thus we can conclude that our proposed MSNN has relatively clearer patterns than the other models; in other words, our method thoroughly represents input EEG signal spatial features.

C. t-SNE Visualization of Learned Features

To validate the representation ability of the proposed network, we plotted t-SNE transformed learned features shown in Fig. 5(b). Specifically, we exhibited extracted features from test SSVEP EEG samples from the first, second, and third spatio-spectral-temporal feature representation layers, *i.e.*, $\mathbf{f}_1^{\text{SST}}$, $\mathbf{f}_2^{\text{SST}}$, and $\mathbf{f}_3^{\text{SST}}$ (first three figures in Fig. 5(b)). Then, we also depicted the final learned feature, *i.e.*, $\mathcal{G}(\mathbf{f}_{\text{concat}}^{\text{SST}})$. These intermediate features $\mathbf{f}_1^{\text{SST}}$, $\mathbf{f}_2^{\text{SST}}$, and $\mathbf{f}_3^{\text{SST}}$ are temporally pooled just for visualization like $\mathcal{G}(\mathbf{f}_{\text{concat}}^{\text{SST}})$. We used the first subject's

¹³See 'Supplementary F: Additional Results for Activation Pattern' for additional results of the activation patterns and analyses in the quantitative standpoint.

From this phenomenon, we can conclude that our proposed MSNN can capture important features on the multi-scale range, not only in the frequency of interest.

first session data in the KU-SSVEP dataset [8], and used a learning rate of 200, a perplexity of 10 for the t-SNE calculation, and visualization.

From these visualized represented features, we could observe that $\mathcal{G}(\mathbf{f}_{\text{concat}}^{\text{SST}})$ is more class-discriminative than the other intermediate features. Additionally, we observed a trend, which demonstrated that a feature learned by a deeper layer is more disentangled than others learned by shallower layers.

D. Mental Fatigue Classification

For the application analysis of drowsiness detection, we visualized confusion matrices that were estimated by the experimental results of the SEED-VIG dataset [36] in Fig. 5(c). Since the labels that identify the mental status were decided using the PERCLOS levels [36], the label at the boundary of the two classes may not be accurate. In this respect, we can conclude that the proposed method is useful for drowsiness state detection because false detections predicted by the proposed method are mostly at the boundaries between classes, *e.g.*, the 'awake' *vs.* 'tired' or 'tired' *vs.* 'drowsy' case. In addition, for practical application, it is essential to detect the drowsy state accurately to avoid unexpected situations, such as a car accident. The proposed method achieved the highest and most promising result for detecting drowsiness among other baselines, *i.e.*, it achieved the highest precision score for identifying the drowsy state. Therefore, we can also expect that our proposed method can be applied in real-world situations.

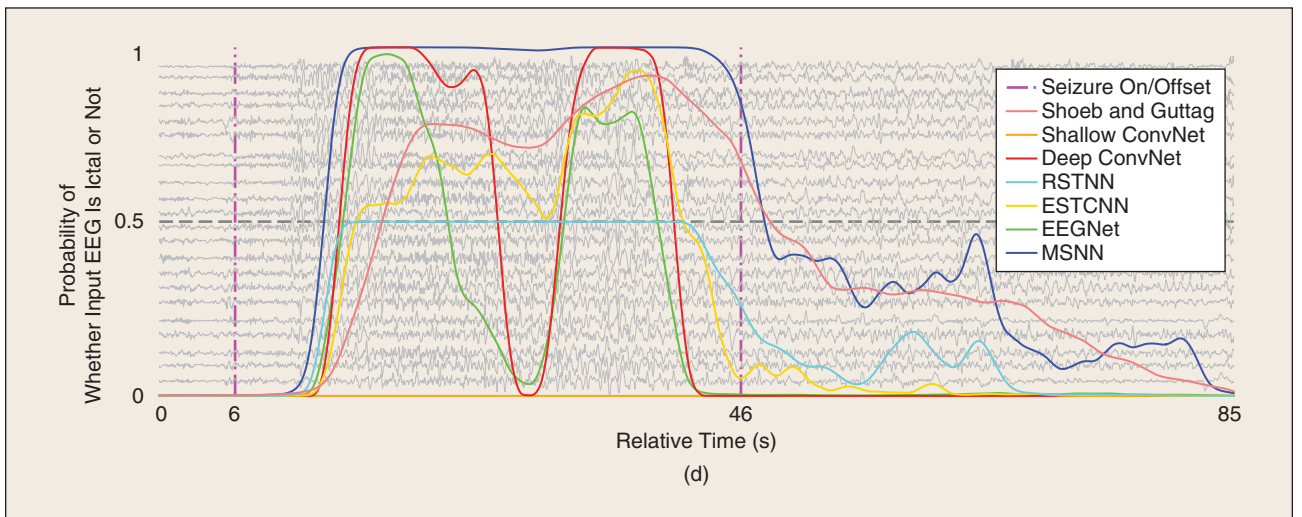


FIGURE 5 (Continued) Investigation of learned weights (Fig. 5(a)) and represented features (Fig. 5(b)), and inspection of the practical usage of the proposed network (Fig. 5(c) and 5(d)). (d) Changes of probabilities estimated by comparable baselines, and the proposed method. These plots demonstrate the probability of whether input EEG is ictal or not. Two dot-dashed lines (magenta) denote the seizure onset and ending, respectively, labeled by Shoeb [12].

E. Early Seizure Detection

Early detection [37] of seizures is one of the most important potential practical applications for this work. Hence, we also validated the benefits of the proposed method in early seizure detection. Specifically, in the training phase, the MSNN was trained using normal and ictal EEG samples with binary labels (e.g., 0: normal and 1: seizure) similar to a conventional training framework. In the testing phase, we input the EEG samples using a sliding window with a 1/256 stride. Then, we observed the change in the output probability values to determine the character of the input (normal or ictal).

Additionally, we visualized these changes in Fig. 5(d) (We used the first subject's third EEG trial in the CHB-MIT dataset [12] for the visualization). In Fig. 5(d), magenta-colored dot-dashed lines denote the seizure onset and offset. Colored solid lines denote the probability change of various methods. In this visualization, we observed that the proposed method is more stable for detecting seizures. Specifically, the proposed method detects the seizure EEG signal as a seizure state with a strong probability (almost 1), whereas the other methods have low confidence values (Shoeb and Guttag [14]'s method and EST-CNN [26]) or even make incorrect decisions regarding the seizure state (Shallow ConvNet [3], Deep ConvNet [3], RSTNN [18], and EEGNet [4]).

F. Towards EEG Representation Method

In our work, a distinguished strength of the proposed MSNN is a reasonable discriminative ability in various paradigms. It is worth noting that this strength can be effectively used in *newly-designed* BCI protocols or paradigms for which we do not have any existing *reasonably-empowered* models for classification or known neurophysiological background useful for model building. Additionally, to suppress heuristic approaches in the studies, combining the proposed network with automated machine learning techniques [43] can maximize its usefulness.

To show this, we designed and conducted additional experiments. First, we used our MSNN for a P300-based online home application control BCI system [44]. The P300 dataset consists of two different tasks: target and non-target P300 that

are acquired from 30 subjects, recorded from 31 electrode channels according to the standard 10–20 system. In the experiment, the MSNN was trained with raw EEG signals of 10 repetitions. As summarized in Table IV, the proposed network outperformed the baseline proposed by the original work [44]. Second, we also used the MSNN for an imagined speech EEG classification task. The imagined speech paradigm lacks neurophysiological finds, thus there is not enough neurophysiological prior knowledge [45]. We exploited BCI Competition V-3 dataset¹⁴, which consists of five different imagined speech: 'hello,' 'help me,' 'stop,' 'thank you,' and 'yes' that are acquired from 15 subjects, recorded from 64 electrode channels according to the standard 10–20 system. In this experiment, our proposed network was also trained with raw imagined speech EEG samples. As summarized in Table V, the proposed network not only outperformed the other baselines but also showed promising performance for five-class imagined speech classification task. Based on these preliminary experimental results, we expect that the MSNN can be regarded as a *starting point* or a *backbone architecture* to a variety of BCI systems including newly-designed protocols and paradigms.

VI. Conclusion

In this work, we proposed a novel and compact deep multi-scale neural network which can learn multi-scale EEG signal features. In our experiments, we validated our novel architecture's effectiveness over diverse EEG paradigms, MI, SSVEP, seizure, and drowsy EEG signals. Even though our proposed MSNN cannot achieve superior performance in every task, it still shows reasonably high performance comparable to the best one without major structural modification. Thus, it is expected that our MSNN can be used as a *general* EEG representation learner for various EEG-based BCI. Furthermore, we inspected the relevance scores to demonstrate the benefits of the multi-scale feature extraction ability, investigated activation pattern maps to understand what types of neurophysiological phenomena were learned by our CNN model, and visualized the t-SNE of learned features to examine the ability of our method to differentiate feature classes. Finally, we also demonstrated that the proposed method can be used for precise drowsiness detection and early seizure detection. In all these respects, we concluded that the proposed deep multi-scale neural network offers significant potential for interpreting EEG signals. Additionally, because the proposed network is clearly *generalizable* to various EEG paradigms, it is expected to bring promising benefits that can be applied to neural architecture search methods [46], thereby making a deep learning-based BCI adaptable to different paradigms.

From a practical standpoint, there are still source limitations with regard to the inter-subject variation [29] in performance or BCIs with multiple classes. In this work, the datasets used in our experiments have a small number of classes, which are definitely not enough for practical use. However, it is important for a BCI system to be robust for multi-class settings. In this regard,

TABLE IV Performance evaluation on the P300 dataset [44]. The Method column denotes a baseline [44] and the proposed method. * denotes evaluation performances taken from [44].

METHOD	MEAN±STD.	MAX	MIN
PCA + SVM [44]	.83±.18*	1.00*	.40*
MSNN (PROPOSED)	.86±.17	1.00	.43

TABLE V Performance evaluation on the imagined speech dataset. The Method row denotes baselines and the proposed method.

METHOD	SHALLOW CONVNET [3]	DEEP CONVNET [3]	EEGNET [4]	MSNN (PROPOSED)
ACCURACY	.51±.07	.64±.12	.63±.08	.67±.09

¹⁴Available at: <https://osf.io/pq7vb/>

we will focus on developing a practical and more plausible multi-class BCI system with the application of adversarial learning [47], [48] or other learning strategies.

Acknowledgment

This work was supported by Institute for Information & Communications Technology Promotion (IITP) grant funded by the Korea government under Grant 2017-0-00451 (Development of BCI based Brain and Cognitive Computing Technology for Recognizing User's Intentions using Deep Learning) and Grant 2019-0-00079 (Department of Artificial Intelligence, Korea University)

This article has supplementary downloadable material available at <https://doi.org/10.1109/MCI.2021.3061875>, provided by the authors.

References

- [1] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, "Optimizing Spatial Filters for Robust EEG Single-trial Analysis," *IEEE Signal Process. Mag.*, vol. 25, no. 1, pp. 41–56, 2008. doi: 10.1109/MSP.2008.4408441.
- [2] Y. Wang, M. Nakanishi, and D. Zhang, "EEG-based brain-computer interfaces," *Adv. Exp. Med. Biol.*, vol. 1101, pp. 41–65, 2019. doi: 10.1515/revneuro.2010.21.6.451.
- [3] R. T. Schirmer et al., "Deep learning with convolutional neural networks for EEG decoding and visualization," *Hum. Brain Mapp.*, vol. 38, no. 11, pp. 5391–5420, 2017. doi: 10.1002/hbm.23730.
- [4] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for eeg-based brain-computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, p. 056013, 2018. doi: 10.1088/1741-2552/aae8c8.
- [5] W. Ko, E. Jeon, and H.-I. Suk, "A novel RL-assisted deep learning framework for task-informative signals selection and classification for spontaneous BCIs," *IEEE Trans. Inf. Inform.*, 2020.
- [6] P. Aricò, G. Borghini, G. Di Flumeri, N. Sciaraffa, and F. Babiloni, "Passive BCI beyond the lab: Current trends and future directions," *Physiol. Meas.*, vol. 39, no. 8, p. 08TR02, 2018. doi: 10.1088/1361-6579/aa57e.
- [7] X. Zhang, L. Yao, X. Wang, J. Monaghan, and D. McAlpine, "A survey on deep learning based brain computer interface: Recent advances and new frontiers," 2019, arXiv:1905.04149.
- [8] M.-H. Lee et al., "EEG Dataset and OpenBMI toolbox for three BCI paradigms: An investigation into BCI illiteracy," *GigaScience*, vol. 8, no. 5, p. giz002, 2019. doi: 10.1093/gigascience/giz002.
- [9] M. Nakanishi, Y. Wang, Y.-T. Wang, and T.-P. Jung, "A comparison study of canonical correlation analysis based methods for detecting steady-state visual evoked potentials," *PLoS ONE*, vol. 10, no. 10, p. e0140703, 2015. doi: 10.1371/journal.pone.0140703.
- [10] J. Jin et al., "The study of generic model set for reducing calibration time in P300-based brain-computer interface," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 1, pp. 3–12, 2019. doi: 10.1109/TNSRE.2019.2956488.
- [11] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, "Filter bank common spatial pattern (FBCSP) in brain-computer interface," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2008, pp. 2390–2397.
- [12] A. H. Shoeb, "Application of machine learning to epileptic seizure onset detection and treatment," Ph.D. dissertation, Massachusetts Inst. Technol., 2009.
- [13] H.-I. Suk and S.-W. Lee, "A novel Bayesian framework for discriminative feature extraction in brain-computer interfaces," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 2, pp. 286–299, 2012. doi: 10.1109/TPAMI.2012.69.
- [14] A. H. Shoeb and J. V. Guttg, "Application of machine learning to epileptic seizure detection," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 975–982.
- [15] A. M. Ray et al., "A subject-independent pattern-based brain-computer interface," *Front. Behav. Neurosci.*, vol. 9, p. 269, 2015. doi: 10.3389/fnbeh.2015.00269.
- [16] F. Lotte, C. Guan, and K. K. Ang, "Comparison of designs towards a subject-independent brain-computer interface based on motor imagery," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2009, pp. 4543–4546. doi: 10.1109/IEMBS.2009.5334126.
- [17] D. Zhang et al., "Cascade and parallel convolutional recurrent neural networks on EEG-based intention recognition for brain computer interface," in *Proc. 32nd AAAI Conf. Artif. Intell. (AAAI)*, 2018.
- [18] W. Ko, J. Yoon, E. Kang, E. Jun, J.-S. Choi, and H.-I. Suk, "Deep recurrent spatio-temporal neural network for motor imagery based BCI," in *Proc. 6th Int. Winter Conf. Brain-Comput. Interface (BCI)*, 2018, pp. 1–3.
- [19] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1251–1258.
- [20] A. Supratak, H. Dong, C. Wu, and Y. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 11, pp. 1998–2008, 2017. doi: 10.1109/TNSRE.2017.2721116.
- [21] S. Sakhavi, C. Guan, and S. Yan, "Learning temporal information for brain-computer interface using convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, 2018. doi: 10.1109/TNNLS.2018.2789927.
- [22] N. Waytowich et al., "Compact convolutional neural networks for classification of asynchronous steady-state visual evoked potentials," *J. Neural Eng.*, vol. 15, no. 6, p. 066031, 2018. doi: 10.1088/1741-2552/aae5d8.
- [23] N.-S. Kwak, K.-R. Müller, and S.-W. Lee, "A convolutional neural network for steady state visual evoked potential classification under ambulatory environment," *PLoS ONE*, vol. 12, no. 2, p. e0172578, 2017. doi: 10.1371/journal.pone.0172578.
- [24] U. Asif, S. Roy, J. Tang, and S. Harrer, "SeizureNet: A deep convolutional neural network for accurate seizure type classification and seizure detection," 2019, arXiv:1903.03232.
- [25] A. Emami, N. Kunii, T. Matsuo, T. Shinozaki, K. Kawai, and H. Takahashi, "Seizure detection by convolutional neural network-based analysis of scalp electroencephalography plot images," *NeuroImage Clin.*, vol. 22, p. 101.684, 2019. doi: 10.1016/j.nicl.2019.101684.
- [26] Z. Gao et al., "EEG-based spatio-temporal convolutional neural network for driver fatigue evaluation," *IEEE Trans. Neural Netw. Learn. Syst.*, 2019. doi: 10.1109/TNNLS.2018.2886414.
- [27] O.-Y. Kwon, M.-H. Lee, C. Guan, and S.-W. Lee, "Subject-independent brain-computer interfaces based on deep convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, 2019. doi: 10.1109/TNNLS.2019.2946869.
- [28] W. Ko, K. Oh, E. Jeon, and H.-I. Suk, "VIGNet: A deep convolutional neural network for EEG-based driver vigilance estimation," in *Proc. 8th Int. Winter Conf. Brain-Comput. Interface (BCI)*, 2020, pp. 1–3. doi: 10.1109/BCI48061.2020.9061668.
- [29] V. Jayaram, M. Alamgir, Y. Altun, B. Scholkopf, and M. Grosse-Wentrup, "Transfer learning in brain-computer interfaces," *IEEE Comput. Intell. Mag.*, vol. 11, no. 1, pp. 20–31, 2016. doi: 10.1109/MCI.2015.2501545.
- [30] S. Haufe et al., "On the interpretation of weight vectors of linear models in multivariate neuroimaging," *NeuroImage*, vol. 87, pp. 96–110, 2014. doi: 10.1016/j.neuroimage.2013.10.067.
- [31] J. Jin, Y. Miao, I. Daly, C. Zuo, D. Hu, and A. Cichocki, "Correlation-based channel selection and regularized feature optimization for MI-based BCI," *Neural Netw.*, vol. 118, pp. 262–270, 2019. doi: 10.1016/j.neunet.2019.07.008.
- [32] J. Jin et al., "Bispectrum-based channel selection for motor imagery based brain-computer interfacing," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 10, pp. 2153–2163, 2020. doi: 10.1109/TNSRE.2020.3020975.
- [33] J. Jin, R. Xiao, I. Daly, Y. Miao, X. Wang, and A. Cichocki, "Internal feature selection method of CSP based on L1-norm and Dempster-Shafer theory," *IEEE Trans. Neural Netw. Learn. Syst.*, 2020. doi: 10.1109/TNNLS.2020.3015505.
- [34] F. Lotte, "Signal processing approaches to minimize or suppress calibration time in oscillator activity-based brain-computer interfaces," *Proc. IEEE*, vol. 103, no. 6, pp. 871–890, 2015. doi: 10.1109/JPROC.2015.2404941.
- [35] T. H. Sanders, M. McCurry, and M. A. Clements, "Sleep stage classification with cross frequency coupling," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2014, pp. 4579–4582. doi: 10.1109/EMBC.2014.6944643.
- [36] W.-L. Zheng and B.-L. Lu, "A multimodal approach to estimating vigilance using EEG and Forehead EOG," *J. Neural Eng.*, vol. 14, no. 2, p. 026017, 2017. doi: 10.1088/1741-2552/aa5a98.
- [37] J. Lee et al., "Early seizure detection by applying frequency-based algorithm derived from the principal component analysis," *Front. Neuroinform.*, vol. 11, p. 52, 2017. doi: 10.3389/fninf.2017.00052.
- [38] M. Liang and X. Hu, "Recurrent convolutional neural network for object recognition," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 3367–3375. doi: 10.1109/CVPR.2015.7298958.
- [39] M. Lin, Q. Chen, and S. Yan, "Network in Network," 2013, arXiv:1312.4400.
- [40] H. Cho, M. Ahn, S. Ahn, M. Kwon, and S. C. Jun, "EEG datasets for motor imagery brain-computer interface," *GigaScience*, vol. 6, no. 7, p. gix034, 2017. doi: 10.1093/gigascience/gix034.
- [41] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. 13th Int. Conf. Artif. Intell. Statist. (AISTATS)*, 2010, pp. 249–256.
- [42] G. Montavon, S. Lapuschkin, A. Binder, W. Samek, and K.-R. Müller, "Explaining nonlinear classification decisions with deep Taylor decomposition," *Pattern Recognit.*, vol. 65, pp. 211–222, 2017. doi: 10.1016/j.patcog.2016.11.008.
- [43] F. Hutter, L. Kotthoff, and J. Vanschoren, *Automated Machine Learning: Methods, Systems, Challenges*. Springer-Verlag, 2019.
- [44] M. Kim, M.-K. Kim, M. Hwang, H.-Y. Kim, J. Cho, and S.-P. Kim, "Online home appliance control using EEG-based brain-computer interfaces," *Electronics*, vol. 8, no. 10, p. 1101, 2019. doi: 10.3390/electronics8101101.
- [45] S.-H. Lee, M. Lee, and S.-W. Lee, "Neural decoding of imagined speech and visual imagery as intuitive paradigms for BCI communication," *IEEE Trans. Neural Syst. Rehabil. Eng.*, 2020. doi: 10.1109/TNSRE.2020.3040289.
- [46] E. Rapaport, O. Shriki, and R. Puzis, "EEGNAS: Neural architecture search for electroencephalography data analysis and decoding," in *Proc. Int. Work. Hum. Brain Artif. Intell.*, 2019, pp. 3–20. doi: 10.1007/978-981-15-1398-5_1.
- [47] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2096–2030, 2016. doi: 10.5555/2946645.2946704.
- [48] E. Jeon, W. Ko, and H.-I. Suk, "Domain adaptation with source selection for motor-imagery based BCI," in *Proc. 7th Int. Winter Conf. Brain-Comput. Interface (BCI)*, 2019, pp. 1–4. doi: 10.1109/IWW-BCI.2019.8737340.

